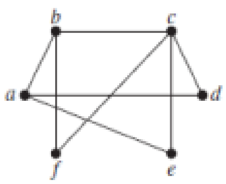
Question 1: 4

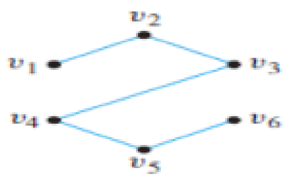
Points

Find which of the following graphs are bipartite. Redraw the bipartite graphs so that their bipartite nature is evident. Also write the disjoint set of vertices.

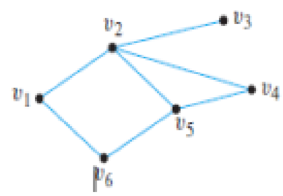
i)



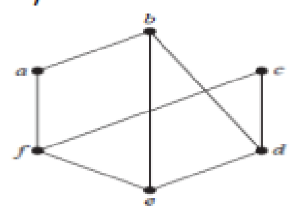
ii)



iii)



iv)



(a) Let $X_n = 2^n + 5.\,3^n$ for $n = 0, 1, 2, 3\ldots$ Show that $X_4 = 5X_3 - 6X_2$

(b) Find the sum of numbers between 250 and 1000 which are divisible by 17.

Question 2: 6 Points

7. Let R be the following relation defined on the set {a, b, c, d}: R = {(a, a), (a, c), (a, d), (b, a), (b, b), (b, c), (b, d), (c, b), (c, c), (d, b), (d, d)}.

Determine whether R is: (a) Reflexive (b) Symmetric (c) Antisymmetric (d) Transitive (e) Irreflexive (f) Asymmetric

a. Draw the directed graph for R1 o R2. Consider the relation R1 = {(4, a), (4, b), (5, c), (6, a), (6, c)} from X to Y and R2 = {(a, l), (a, n), (b, l), (b, m), (c, l), (c, m), (c, n)} from Y to Z where X = {4, 5, 6}, Y = {a, b, c} and Z = {l, m, n}.

) Prove or Disprove that the less than (a<b) relation on Set A = {1, 2, 3, 4} is a partial order relation. Discuss all its properties

Question 3: 2 Points

a) In a group of 15 people, is it possible for each person to have exactly 3 friends? Explain. (Assume that friendship is a symmetric relationship: If x is a friend of y, then y is a friend of x.)

b) In a group of 4 people, is it possible for each person to have exactly 3 friends? Why?

<u>Question 4:</u> **10 Points**

(a) For a given pair of graphs G and G1 in Figure # 1. Determine whether G and G1 are isomorphic. If they are, give function F: V(G) →V(G1) that define the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.

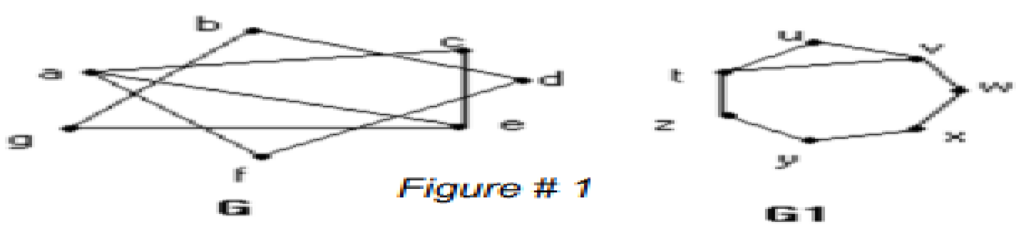
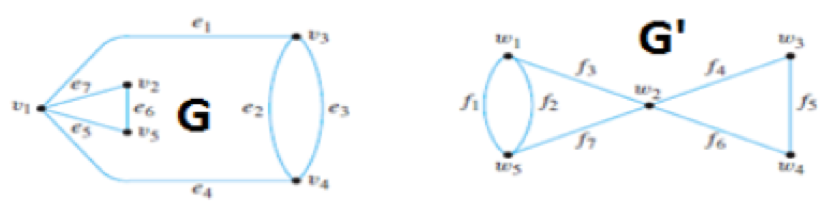


Figure # 1

b.

For given pair (G, G') of graphs. Determine whether they are isomorphic. If they are, give function g: V (G) →V (G') that define the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.

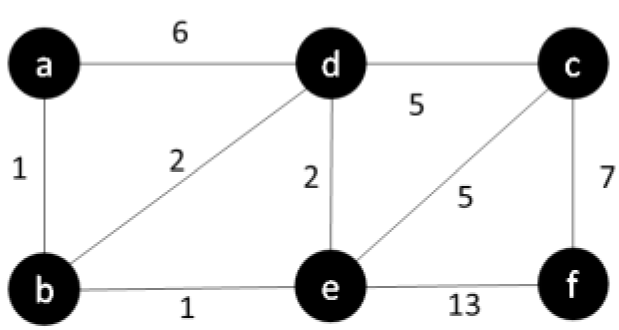
i)



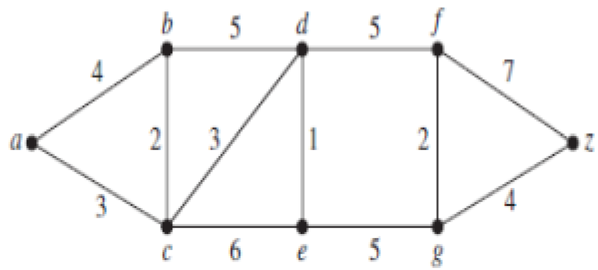
<u>Question 5:</u> **5 Points**

Find the length of a shortest path between *a* and *z* in the given weighted graph by using Dijkstra's algorithm.

1.



2.



<u>**Question 6:**</u>

Determine whether the graph G in Figure # 3 has an Euler circuit OR an Euler path. Construct such a circuit or path when one exists. If no Euler circuit or path exists, justify with a valid reason

Determine whether the graph G in Figure # 3 has a Hamilton circuit OR a Hamilton path. Construct such a circuit or path when one exists. If no Hamilton circuit or path exists, justify with a valid reason.

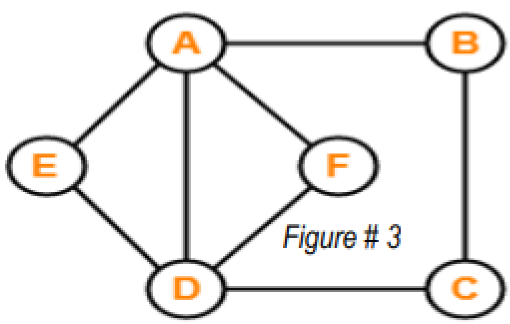


Figure # 3

Use Kruskal's algorithm to find a minimal spanning tree for the graph shown in Figure # 4. Indicate the order in which edges are added to form the tree.

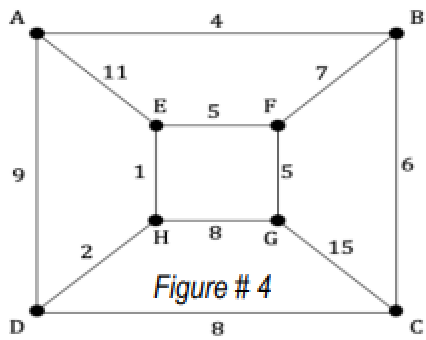
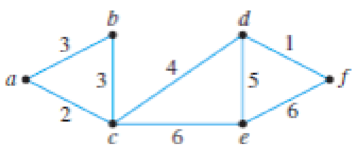


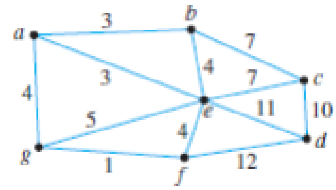
Figure # 4

Use Kruskal's algorithm to find a minimum spanning tree for given graphs. Indicate the order in which edges are added to form each tree.

i)



ii)



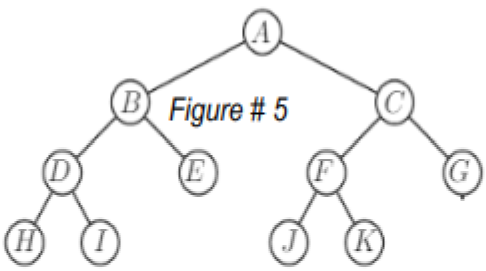


Figure # 5

(c) Show step by step inorder traversal of the tree given in Figure # 5.

(d)

(i) Determine whether the tree given in Figure # 5 is a Full mary tree or not. Give reason.

(ii) Determine whether the tree given in Figure # 5 is a Balanced m-ary tree or not. Give reason.